

Sri Lanka Institute of Information Technology



IT3010 – Network design management Assignment 01

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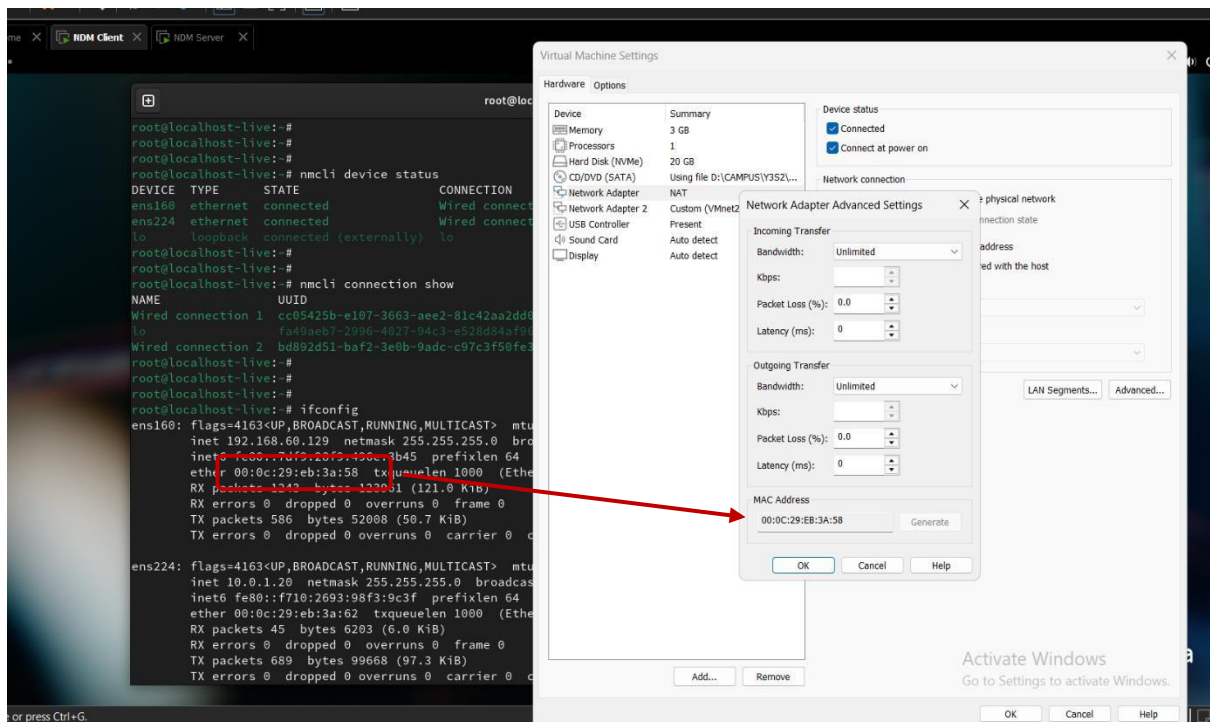
Configure the network settings on fedora client server:

1. Identify network interfaces Commands:

1. nmcli device status
2. nmcli connection show
3. ifconfig

```
root@localhost-live:~#  
root@localhost-live:~#  
root@localhost-live:~#  
root@localhost-live:~# nmcli device status  
DEVICE  TYPE      STATE      CONNECTION  
ens160  ethernet  connected  Wired connection 1  
ens224  ethernet  connected  Wired connection 2  
lo      loopback  connected (externally)  lo  
root@localhost-live:~#  
root@localhost-live:~#  
root@localhost-live:~# nmcli connection show  
NAME                                UUID                                TYPE      DEVICE  
Wired connection 1                  cc05425b-e107-3663-ae2-81c42aa2dd06  ethernet  ens160  
lo                                  fa49aeb7-2996-4027-94c3-e528d84af96b  loopback  lo  
Wired connection 2                  bd892d51-baf2-3e0b-9adc-c97c3f50fe3c  ethernet  ens224  
root@localhost-live:~#  
root@localhost-live:~#  
root@localhost-live:~#  
root@localhost-live:~# ifconfig  
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500  
    inet 192.168.60.129  netmask 255.255.255.0  broadcast 192.168.60.255  
    inet6 fe80::7df9:28f9:496e:3b45  prefixlen 64  scopeid 0x20<link>  
    ether 00:0c:29:eb:3a:58  txqueuelen 1000  (Ethernet)  
    RX packets 1243  bytes 123961 (121.0 KiB)  
    RX errors 0  dropped 0  overruns 0  frame 0  
    TX packets 586  bytes 52008 (50.7 KiB)  
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0  
  
ens224: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500  
    inet 10.0.1.20  netmask 255.255.255.0  broadcast 10.0.1.255  
    inet6 fe80::f710:2693:98f3:9c3f  prefixlen 64  scopeid 0x20<link>  
    ether 00:0c:29:eb:3a:62  txqueuelen 1000  (Ethernet)  
    RX packets 45  bytes 6203 (6.0 KiB)  
    RX errors 0  dropped 0  overruns 0  frame 0  
    TX packets 689  bytes 99668 (97.3 KiB)  
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0
```

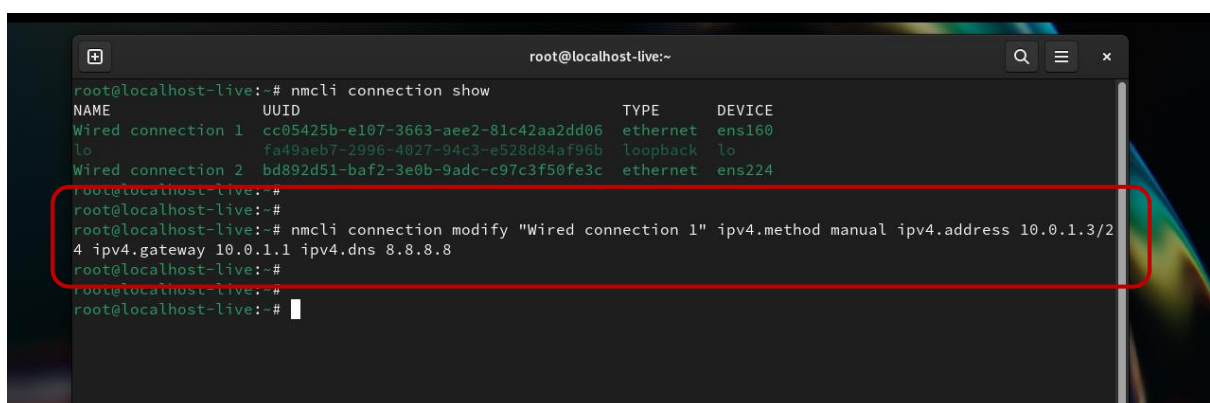
2. Check the mac address of Vmnet2
Go to settings and select network adapter (NMnet2) make it to advance



3. Set static IP for VMnet2 (Wired connection 1)

Commands:

1. `nmcli connection modify "Wired connection 1" ipv4.method manual ipv4.addresses 10.0.1.3/24 ipv4.gateway 10.0.1.1 ipv4.dns 8.8.8.8`



4. Apply the Changes

Commands:

1. `nmcli connection down "Wired connection 1"` 2) `nmcli connection up "Wired connection 1"`

```
Feb 18 14:04
root@localhost-live:~
root@localhost-live:~# nmcli connection show
NAME                UUID                                  TYPE      DEVICE
Wired connection 1  cc05425b-e107-3663-ae2-81c42aa2dd06 ethernet  ens160
lo                  fa49aeb7-2996-4027-94c3-e528d84af96b loopback   lo
Wired connection 2  bd892d51-baf2-3e0b-9adc-c97c3f50fe3c ethernet  ens224
root@localhost-live:~#
root@localhost-live:~# nmcli connection modify "Wired connection 1" ipv4.method manual ipv4.address 10.0.1.3/24
root@localhost-live:~# nmcli connection down "Wired connection 1"
Connection 'Wired connection 1' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/2)
root@localhost-live:~# nmcli connection up "Wired connection 1"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/28)
root@localhost-live:~#
```

5. Check IP

Commands:

1. ip addr / inet 10.0.1.3/24
2. ip route/

```
root@localhost-live:~# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:eb:3a:58 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 10.0.1.3/24 brd 10.0.1.255 scope global noprefixroute ens160
        valid_lft forever preferred_lft forever
    inet6 fe80::7df9:28f9:496e:3b45/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens224: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:eb:3a:62 brd ff:ff:ff:ff:ff:ff
    altname enp19s0
    inet 10.0.1.20/24 brd 10.0.1.255 scope global dynamic noprefixroute ens224
        valid_lft 342sec preferred_lft 342sec
    inet6 fe80::f710:2693:98f3:9c3f/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
root@localhost-live:~#
```

```
valid_lft forever preferred_lft forever
root@localhost-live:~#
root@localhost-live:~#
root@localhost-live:~#
root@localhost-live:~#
root@localhost-live:~# ip route
default via 10.0.1.1 dev ens224 proto dhcp src 10.0.1.20 metric 20101
default via 10.0.1.1 dev ens160 proto static metric 20102
10.0.1.0/24 dev ens224 proto kernel scope link src 10.0.1.20 metric 101
10.0.1.0/24 dev ens160 proto kernel scope link src 10.0.1.3 metric 102
root@localhost-live:~#
root@localhost-live:~#
root@localhost-live:~#
root@localhost-live:~#
root@localhost-live:~#
```

6. Checking Network Connectivity using the ping command

Commands:

1. ping 10.0.1.2
2. ping 10.0.1.3

```
root@localhost-live:~#
root@localhost-live:~#
root@localhost-live:~# ping 10.0.1.3
PING 10.0.1.3 (10.0.1.3) 56(84) bytes of data.
64 bytes from 10.0.1.3: icmp_seq=1 ttl=64 time=4.40 ms
64 bytes from 10.0.1.3: icmp_seq=2 ttl=64 time=3.41 ms
64 bytes from 10.0.1.3: icmp_seq=3 ttl=64 time=0.118 ms
64 bytes from 10.0.1.3: icmp_seq=4 ttl=64 time=0.097 ms
64 bytes from 10.0.1.3: icmp_seq=5 ttl=64 time=0.129 ms
64 bytes from 10.0.1.3: icmp_seq=6 ttl=64 time=0.122 ms
^C
--- 10.0.1.3 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5549ms
rtt min/avg/max/mdev = 0.097/1.379/4.402/1.809 ms
root@localhost-live:~#
root@localhost-live:~# ping 10.0.1.2
PING 10.0.1.2 (10.0.1.2) 56(84) bytes of data.
64 bytes from 10.0.1.2: icmp_seq=1 ttl=64 time=4.22 ms
64 bytes from 10.0.1.2: icmp_seq=2 ttl=64 time=3.09 ms
64 bytes from 10.0.1.2: icmp_seq=3 ttl=64 time=1.16 ms
64 bytes from 10.0.1.2: icmp_seq=4 ttl=64 time=4.58 ms
64 bytes from 10.0.1.2: icmp_seq=5 ttl=64 time=4.48 ms
64 bytes from 10.0.1.2: icmp_seq=6 ttl=64 time=4.35 ms
^C
--- 10.0.1.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5463ms
rtt min/avg/max/mdev = 1.162/3.646/4.576/1.215 ms
root@localhost-live:~#
```